



Recap

Rules

product rule $p(a,b) = p(b|a) p(a)$

Sum/marginalization rule $p(b) = \sum_a p(a,b)$
 $\int p(a,b) da$

Conditional probability $p(a|b) = \frac{p(a,b)}{p(b)}$

conditional independence $p(a,b,c) = p(a|c)p(b|c)$

Intuition

split a joint event (a,b) into "first a , then b "

average out or marginalize a variable you don't care about

update belief about a after observing b

Once you know c , learning a tells you nothing more about b

mini example

- draw an ace and then a king w/out replacement
 $p(\text{King, Ace}) = p(\text{King} | \text{Ace}) p(\text{Ace})$
 $= \frac{4}{51} \cdot \frac{4}{52} = \frac{16}{2652}$

Overall chance of sock being red = $\sum$ over both drawers. $p(D1)=0.5$, $p(\text{red} | D1) = \frac{1}{2}$, $p(\text{red} | D2) = \frac{1}{4}$, $p(\text{red}) = 0.5 \cdot \frac{1}{2} + 0.5 \cdot \frac{1}{4} = \frac{3}{8}$

$p(\text{spam}) = 0.3$; $p(\text{spam}, \text{"free"}) = 0.2$, then we see cond. prob from $p(\text{"free"} | \text{spam}) = \frac{p(\text{"free", spam})}{p(\text{spam})} = \frac{0.2}{0.3} = 0.67$

$P(R) = \text{raining}$; $P(T) = \text{traffic}$; $P(S) = \text{sun/day}$ $\rightarrow$ roads are empty on Sun
 $P(R|S) = 0.2$; $P(T|S) = 0.1$; $p(R|S) \cdot p(T|S) = 0.2 \cdot 0.1 = 0.02$

- given some data y , what can we say about a new observation y^* ?

- step 1: formulate joint distribution $\forall$ variables of interest.

$$p(y^*, y, \theta) = p(y^*, y, \theta) p(\theta) = p(y^* | \theta) p(y | \theta) p(\theta)$$

Product rule: $p(a,b) = p(b|a) p(a)$

- step 2: Condition on the observed data y

$$p(y^*, \theta | y) = \frac{p(y^*, y, \theta)}{p(y)} = \frac{p(y^* | \theta) p(y | \theta) p(\theta)}{p(y)}$$

- step 3: marginalize out parameter θ using the sum rule to get the posterior predictive distribution

$$p(y^* | y) = \int p(y^*, \theta | y) d\theta = \int \frac{p(y^* | \theta) p(y | \theta) p(\theta)}{p(y)} d\theta = \int p(y^* | \theta) p(\theta | y) d\theta = \mathbb{E}_{p(\theta | y)} [p(y^* | \theta)]$$

predictive likelihood
likelihood for y^*

average w.r.t to $p(\theta | y)$

Bayesian inference for supervised learning

$$p(w | y) = \frac{p(y | w) p(w)}{p(y)}$$

Compute posterior $p(w | y)$

Compute predictive $p(y^* | x^*, y)$

$$p(y^* | y) = \int p(y^* | x^*, w) p(w | y) dw$$

Summarize using mean, mode, initials

$$D = \{x_1, y_1, \dots, x_n, y_n\}$$

$$p(w, y) = p(y | w) p(w)$$

Linear Regression: the probabilistic model

- w.r.t. $\mathcal{F}$

$$p(y^* | D, x)$$

(the prob. of future target y^* given all observed data $D = \{(x_i, y_i)\}_{i=1}^N$ and new input x^*)
 a single pt. prediction has no notion of risk. The distribution tells us both what we expect a linear function may

• Signal model (deterministic part)

$$f(x_i | w) = w^T \phi(x_i)$$

new input or pre-processed
 use our linear in the weights
 vector basis functions (could be $[1, x]$ for straight line) or $[1, x, x^2]$ for quadratic

• noise model (stochastic part)

$$y_i = f(x_i | w) + \epsilon_i, \quad \epsilon_i \sim \mathcal{N}(0, \sigma^2)$$

- additive noise (Gaussian) captures everything the deterministic function can't explain

• Likelihood of one data point

$$p(y_i | x_i, w, \sigma^2) = \mathcal{N}(y_i | w^T \phi(x_i), \sigma^2)$$

- given a guess of w and σ^2 , each point's probability density is a Gaussian centered at the model's prediction, with width σ

- for the whole dataset

$$p(y | X, w, \sigma^2) = \prod_{i=1}^N \mathcal{N}(y_i | w^T \phi(x_i), \sigma^2)$$

• MLE

1) Log likelihood

$$\log p(y | X, w, \sigma^2) = -\frac{N}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^N (y_i - w^T \phi(x_i))^2$$

- closed form solution let $\Phi \in \mathbb{R}^{N \times n}$ be a design matrix where i -th row is $\phi(x_i)^T$ and $y = (y_1, \dots, y_N)^T$. Then

$$\hat{w}_{MLR} = (\Phi^T \Phi)^{-1} \Phi^T y$$

Ordinary least squares

$$\sigma_{MLR}^2 = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{w}_{MLR}^T \phi(x_i))^2$$

• plug-in predictive distribution

$$p(y^* | D, x) \approx \mathcal{N}(y^* | \hat{w}_{MLR}^T \phi(x^*), \sigma^2)$$

plug-in constants

Example

- we use the following $\phi(x) = \begin{bmatrix} 1 \\ x \end{bmatrix}$, $f(x|w) = w^T \phi(x) = w_0 + w_1 x$, and iid noise $\epsilon_i \sim \mathcal{N}(0, \sigma^2)$

i	x_i	y_i
1	0	1.10
2	1	2.90
3	2	5.20

1) we build design matrix $\Phi = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}$, $y = \begin{bmatrix} 1.10 \\ 2.90 \\ 5.20 \end{bmatrix}$

2) compute w_{MLE}

2.1) the normal equation matrix $\Phi^T \Phi = \begin{bmatrix} 3 & 3 \\ 3 & 5 \end{bmatrix}$

2.2) the inverse, determinant = $3 \cdot 5 - 3 \cdot 3 = 6$

$$(\Phi^T \Phi)^{-1} = \frac{1}{6} \begin{bmatrix} 5 & -3 \\ -3 & 3 \end{bmatrix} = \begin{bmatrix} \frac{5}{6} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

2.3) the RHS

$$\Phi^T y = \begin{bmatrix} 1.10 + 2.90 + 5.20 \\ 0 \cdot 1.10 + 1 \cdot 2.90 + 2 \cdot 5.20 \end{bmatrix} = \begin{bmatrix} 9.20 \\ 19.30 \end{bmatrix}$$

2.4) Multiply to get w_{MLE}

$$w_{MLE} = (\Phi^T \Phi)^{-1} \Phi^T y = \begin{bmatrix} 1.02 \\ 2.05 \end{bmatrix}$$

3. Estimate the noise variance σ^2

3.1 fitted values $\hat{y} = \Phi w_{MLE} = [1.02, 3.07, 5.12]^T$

3.2 Residuals, $r = y - \hat{y} = [0.08, -0.17, 0.08]^T$

3.3 Plug-in estimator $\hat{\sigma}^2 = \frac{1}{N} \sum_{i=1}^N r_i^2 = \frac{0.08^2 + (-0.17)^2 + 0.08^2}{3} \approx \underline{\underline{0.0139}}$

so $\hat{\sigma} \approx 0.12$

4. Predict at a new input $x^* = 1.5$

4.1 Basis vector $\phi(1.5) = \begin{bmatrix} 1 \\ 1.5 \end{bmatrix}$

4.2 Predictive mean $\mu^* = w_{MLE}^T \phi(1.5) = 1.02 + 2.05 \cdot 1.5 = 4.10$

$\therefore p(y^* | D, x^* = 1.5) \sim \mathcal{N}(4.10, 0.0139)$ the estimate of σ^2

Bayesian Linear Regression

- σ^2 is fixed and known

- we start from bayes rule

$$p(w|y) = \frac{p(y|w) p(w)}{p(y)}$$

Posterior Predictive Distribution

- The PPD is given by

$$\begin{aligned}
 p(y^* | y, x^*) &= \int \underbrace{p(y^* | x^*, w)}_{\text{pred lik}} \underbrace{p(w | y)}_{\text{posterior}} dw \\
 &= \int \mathcal{N}(y^* | w^T \phi(x^*), \sigma^2) \mathcal{N}(w | m, S) dw \\
 &= \int \mathcal{N}(y^* | (m^T \phi(x^*))^T, \sigma^2 + \phi(x^*)^T S \phi(x^*)) \\
 &= \int \mathcal{N}(y^* | \phi(x^*)^T m, \sigma^2 + \phi(x^*)^T S \phi(x^*))
 \end{aligned}$$

using Murphy (3.38) $p(y) = \int \mathcal{N}(z | u_2, \Sigma_2) \mathcal{N}(y | w_2 + b, \Sigma_y) dz = \mathcal{N}(y | w_{u_2} + b, \Sigma_y + W \Sigma_2 W^T)$

Task 1.1 Compute MLE, MAP & Posterior Mean

- we have a following setting:

- $N=20$

- $x = \begin{bmatrix} x_1 \\ \vdots \\ x_{20} \end{bmatrix}$

- $y = \begin{bmatrix} y_1 \\ \vdots \\ y_{20} \end{bmatrix}$

- $\alpha = 5$ (Prior precision for slope and intercept)

- $\beta = \frac{3}{4}$ (Likelihood precision - i.e. inverse of observation noise)

- prior $p(w) = \mathcal{N}(w | 0, \alpha^{-1} I_2) = \mathcal{N}(w | 0, \frac{1}{5} I_2)$

- likelihood $p(y | x, w, \sigma^2) = \prod_{n=1}^N \mathcal{N}(y_n | w^T \phi(x_n), \sigma^2) = \mathcal{N}(y | \Phi w, \sigma^2 I_2) = \mathcal{N}(y | \Phi w, \frac{1}{\beta} I) = \mathcal{N}(y | \Phi w, \frac{1}{\frac{3}{4}} I)$

- model: $f(x | w) = \phi_n^T w$, where $\phi_n = \begin{bmatrix} 1 \\ x_n \end{bmatrix}$

- design matrix $\Phi = \begin{bmatrix} 1 & x_1 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix}$, where $\Phi \in \mathbb{R}^{N \times D}$, with $D=2$

- calculate $\hat{w}_{MLE} = (\Phi^T \Phi)^{-1} \Phi^T y$

$$= \begin{bmatrix} 0.5 \\ 1.7 \end{bmatrix}$$

- calculate $\hat{w}_{MAP} = (\beta \Phi^T \Phi + \alpha I)^{-1} \Phi^T y$

$$= \begin{bmatrix} 0.54 \\ 1.24 \end{bmatrix}$$

- calculate posterior mean $m = \beta \Phi^T y$

$$= \begin{bmatrix} 0.54 \\ 1.24 \end{bmatrix}$$

Task 1.2 what is the role of α & β

- the parameters α & β enter the posterior through

controls the diagonal terms of

$$\alpha I = \begin{bmatrix} \alpha & 0 \\ 0 & \alpha \end{bmatrix}$$

$$\bar{S}^{-1} = \underbrace{\alpha I}_{\text{Prior influence}} + \underbrace{\beta \Phi^T \Phi}_{\text{data}}$$

$$m = \beta \Phi^T y$$

α (prior strength)

- acts like a spring pulling weights back to 0.

- bigger $\alpha \Rightarrow$ heavier spring $\Rightarrow$ posterior sticks closer to origin, further from the MLE

β (data confidence)

scales how much curvature the data contributes

Bigger $\beta \Rightarrow$ "clean" data $\Rightarrow$ likelihood is sharp $\Rightarrow$ posterior hugs the likelihood and moves towards it

task 1.3 what happens to the posterior mean m $\wedge$ posterior covariance S $\beta \rightarrow 0$

from eq, we see that the posterior covariance reverts to prior covariance, when $\beta \rightarrow 0$

$$\lim_{\beta \rightarrow 0} S = \lim_{\beta \rightarrow 0} (S_0^{-1} + \beta \Phi^T \Phi)^{-1} \stackrel{S_0 = \alpha I}{=} (S_0^{-1})^{-1} = S_0 = \alpha I$$

Similarly for posterior mean

$$m = \beta S \Phi^T y \quad \text{-- substitute } S = (S_0^{-1} + \beta \Phi^T \Phi)^{-1} \text{ into the mean}$$

$$m = \beta (S_0^{-1} + \beta \Phi^T \Phi)^{-1} \Phi^T y \quad \text{we notice: } S_0^{-1} + \beta \Phi^T \Phi = \beta (\beta^{-1} S_0^{-1} + \Phi^T \Phi)$$

$$\text{so, } (S_0^{-1} + \beta \Phi^T \Phi)^{-1} = (\beta (\beta^{-1} S_0^{-1} + \Phi^T \Phi))^{-1} = \beta^{-1} (\beta^{-1} S_0^{-1} + \Phi^T \Phi)^{-1}$$

-- substitute that back in

$$m = \beta \beta^{-1} (\beta^{-1} S_0^{-1} + \Phi^T \Phi)^{-1} \Phi^T y$$

$$m = (\beta^{-1} S_0^{-1} + \Phi^T \Phi)^{-1} \Phi^T y$$

$$\text{now } \lim_{\beta \rightarrow 0} m = \lim_{\beta \rightarrow 0} (\underbrace{\beta^{-1} S_0^{-1}}_{\text{this term dominates}} + \Phi^T \Phi)^{-1} \Phi^T y$$

$$m \approx (\beta^{-1} S_0^{-1})^{-1} \Phi^T y = \beta S_0 \Phi^T y \rightarrow 0$$

m converges to 0 which is the prior mean, as $\beta \rightarrow 0$. Since β controls the precision of our observations, it makes sense that the model becomes indep. of the data when the precision goes to 0.

task 1.4 Show when we make the prior distribution infinitely wide ($\alpha \rightarrow 0$), then posterior mean converges to the MLE

hint: $S_0 = \alpha^{-1} I$

o Substitute the expression for the posterior covariance into the expression for the posterior mean and take the limit

$$\begin{aligned} m &= \beta S \Phi^T y = \beta (\alpha I + \beta \Phi^T \Phi)^{-1} \Phi^T y \Rightarrow \lim_{\alpha \rightarrow 0} \beta (\alpha I + \beta \Phi^T \Phi)^{-1} \Phi^T y \\ &\Rightarrow \beta (\beta \Phi^T \Phi)^{-1} \Phi^T y \\ &= \beta \beta^{-1} (\Phi^T \Phi)^{-1} \Phi^T y \\ &= (\Phi^T \Phi)^{-1} \Phi^T y \\ &= \hat{w}_{MLE} \end{aligned}$$

Example

- we have a simple model

$$y_n = a + bx_n + e_n = [1 \ x_n] \begin{bmatrix} a \\ b \end{bmatrix} + e_n = \phi_n^T W + e_n$$

intercept
slope
 $\phi(x_n) = \begin{bmatrix} 1 \\ x_n \end{bmatrix}$

- we assume zero-mean and isotropic Gaussian prior,

$m_0 = 0$ mean
 $S_0 = \alpha^{-1} I$ covariance

$\Phi = [\phi(x_1), \dots, \phi(x_N)]^T \in \mathbb{R}^{N \times 2}$ design matrix

$e_n \sim \mathcal{N}(0, \sigma^2)$ homoscedastic noise (precision $\beta = \sigma^{-2}$)

Prior $p(W) = \mathcal{N}(0, \alpha^{-1} I_2)$ isotropic w/ precision α

Closed form posterior

for a Gaussian prior & likelihood

$$S_N^{-1} = \alpha I_2 + \beta \Phi^T \Phi$$

$$m_N = \beta S_N \Phi^T y$$

$$p(W|D) = \mathcal{N}(m_N, S_N)$$

Background

multivariate normal distribution over two variables

$W \sim \mathcal{N}(m_W, S_W)$, where $W = \begin{bmatrix} a \\ b \end{bmatrix}$

$$m_W = \begin{bmatrix} m_{W,1} \\ m_{W,2} \end{bmatrix}$$

$$S_W = \begin{bmatrix} S_{W,11} & S_{W,12} \\ S_{W,21} & S_{W,22} \end{bmatrix}$$

the marginal is

$$a|D \sim \mathcal{N}(m_{N,1}, S_{N,11})$$

$$b|D \sim \mathcal{N}(m_{N,2}, S_{N,22})$$

95% credible intervals for Parameters

$$m_{N,i} \pm 1.96 \sqrt{S_{N,ii}}$$

Credible intervals

- because every marginal of multivariate normal is normal

$$a|D \sim \mathcal{N}(m_{N,1}, S_{N,11})$$

$$b|D \sim \mathcal{N}(m_{N,2}, S_{N,22})$$

$$95\% \text{ cred. int. is: } m_{N,i} \pm 1.96 \sqrt{S_{N,ii}}$$

Predictive (posterior-predictive) distribution

- for a new input x^*

$$p(y^* | x^*, D) = \mathcal{N}(\phi(x^*)^T m_N, \sigma^2 + \phi(x^*)^T S_N \phi(x^*))$$

posterior mean posterior variance

$$\hat{y}_* = 1.96 \sqrt{\sigma^2 + \phi(x_*)^T S_0 \phi(x_*)}$$

Worked out example

h	x_n	y_n
1	1.764	-0.464
2	0.600	2.024
3	0.979	3.191

design matrix

$$\Phi = \begin{bmatrix} 1 & 1.764 \\ 1 & 0.600 \\ 1 & 0.979 \end{bmatrix}, \quad \Phi^T \Phi = \begin{bmatrix} 3 & 3.143 \\ 3.143 & 4.318 \end{bmatrix}$$

hyper parameters

prior precision $\alpha = 1$ (prior variance = 1)

noise variance $\sigma^2 = 1 \rightarrow$ precision $\beta = 1$

Posterior precision and covariance

$$S_N^{-1} = \alpha I_2 + \beta \Phi^T \Phi = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 3 & 3.143 \\ 3.143 & 4.318 \end{bmatrix} = \begin{bmatrix} 4 & 3.143 \\ 3.143 & 4.318 \end{bmatrix}, \quad S_N = (S_N^{-1})^{-1} = \begin{bmatrix} 0.474 & -0.285 \\ -0.285 & 0.362 \end{bmatrix}$$

$\text{var}(a) = 0.474$

$$m_N = \beta S_N \Phi^T y = \frac{1}{1} \begin{bmatrix} 0.474 & -0.285 \\ -0.285 & 0.362 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 1.764 & 0.600 & 0.979 \end{bmatrix} \begin{bmatrix} -0.464 \\ 2.024 \\ 3.191 \end{bmatrix} = \begin{bmatrix} 1.364 \\ -0.224 \end{bmatrix}$$

$\text{var}(b) = 0.362$

Interpretation of the posterior mean

$$y \approx 1.36 - 0.22x$$

a = intercept
slope

95% credible intervals for a and b

- $a \in [1.36 \pm 1.96 \sqrt{0.477}] = [0.01, 2.71]$
- $b \in [-0.22 \pm 1.96 \sqrt{0.362}] = [-1.40, 0.96]$

- we notice slope is very wide (we have change in sign) $\Rightarrow$ meaning we are uncertain

Predictive example of the latent function f^*

- let's predict at $x^* = 1.0$

$$\phi^* = \begin{bmatrix} 1 \\ 1.0 \end{bmatrix}$$

- mean: $\mu_{f^*} = \phi^T(x^*) \cdot m = (1)(1.364) + (1.0)(-0.224) = 1.140$
- variance: $\sigma_{f^*}^2 = \phi^T(x^*) S \phi(x^*)$
 $= \begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} 0.474 & -0.285 \\ -0.285 & 0.362 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$
 $= \phi_1^2 \cdot S_{11} + 2\phi_1\phi_2 S_{12} + \phi_2^2 S_{22}$
 $= 0.266$

Hence, $f^* | y, x \sim \mathcal{N}(1.140, 0.266)$

- credibility intervals: A 95% credible interval is $1.140 \pm 1.96 \sqrt{0.266} = 1.140 \pm 1.01 \Rightarrow (0.13, 2.16)$

Posterior predictive distribution of the observable y^*

- add the measurement-noise variance $\sigma^2 = 1$
- mean is the same
- variance: $\sigma_{y^*}^2 = 0.266 + 1 = 1.266$

so, $y^* | y, x \sim \mathcal{N}(1.140, 1.266)$

- credibility intervals: A 95% predictive interval is $1.140 \pm 1.96 \sqrt{1.266} = 1.140 \pm 2.21 \Rightarrow (-1.07, 3.35)$

implement $p(f^* | x_*, y) \propto p(y_* | x_*, y)$ to compute μ and Σ for a design matrix of Φ^* of new inputs

al OLS we choose a single weight vector $\hat{w}_{OLS} = (\Phi^T \Phi)^{-1} \Phi^T y$, treating it as the truth, and uncertainty is in noise variance.

we have a distribution over possible weights in the prior $p(w) = \mathcal{N}(0, \alpha^{-1} I)$, where we encode a belief, that our w are small and controlled by precision α .

likelihood $p(y | w) = \mathcal{N}(\Phi w, \beta^{-1} I)$, says that the data are a noisy linear projection of the weights

precision β

$$p(w, y) = \mathcal{N}(m, S)$$

$$S^{-1} = \alpha I + \beta \Phi^T \Phi$$

$$m = \beta S \Phi^T y$$

active distributions

ion values $f_* = \phi_*^T w$

this is what the model says before adding measurement noise

observed targets $y_* = f_* + \varepsilon$

this is what we see in the real world (extra noise)

we kept a distribution over w , both f_* and y_* are random.

$$p(f_* | y, \phi_*) = \mathcal{N}(\underbrace{\phi_*^T m}_{\text{mean } \mu_f}, \underbrace{\phi_*^T S \phi_*}_{\text{variance } \sigma_f^2})$$

$$p(y_* | y, \phi_*) = \mathcal{N}(\mu_f, \sigma_f^2 + \beta^{-1})$$

model uncertainty

irreducible noise

Task 3.2 what happens to prior, likelihood, posterior as the nr. of samples N increase?

- **prior**: $p(w) = \mathcal{N}(0, \alpha^{-1}I)$ - we observe that prior keeps unchanged through $N=0$ to $N=20$
- **Likelihood**: $p(y|w) = \mathcal{N}(\Phi w, \beta^{-1}I)$ - we observe that initially when $N=0$, we have no data. However as we increase N the contours gets concentrated around the $\hat{w}_{ML}$
- **Posterior**: $p(w|y)$
 - mean moves towards $\hat{w}_{ML}$
 - when N becomes large, $\alpha^{-1}I$ becomes negligible
 - $m \approx (\Phi^T \Phi)^{-1} \Phi^T y \Rightarrow \hat{w}_{ML}$
 - Covariance Shrinks roughly by $1/N$. Priors influence dies off as the likelihood overwhelms it.
- **Predictions**:
 - predictive uncertainty splits into two pieces $\underbrace{\phi_*^T S \phi_*^{-1}}_{\text{epistemic (}\frac{1}{N}\text{)}} + \underbrace{\beta^{-1}}_{\text{aleatoric (fixed)}}$
 - the green band (latent) tightens to zero width, while the yellow band (observations) bottoms out at the model's noise variance β^{-1}
 - yellow observation band reminds us that $\exists$ world noise you can't reduce regardless the N of data.
 - green (latent) band visualizes the model doubt.

Task 3.3 What happens if you keep $\beta = \frac{3}{4}$ and increase α ?

- **Prior** $p(w) = \mathcal{N}(w | 0, \alpha^{-1}I)$
 - taking variance of $p(w)$, we obtain: $\mathbb{V}[w] = (\alpha^{-1}I) = \alpha^{-1}$ as we increase α as is asked $\frac{1}{\alpha}$ it will get smaller
 - Large α tells us that the weights must be close to 0.
 - precision is likely the reciprocal of variance
precision = α , variance = α^{-1}
 - if you increase α (make the Gaussian tighter), its reciprocal α^{-1} decreases making the cone narrower.
 - Smaller α means a broader prior (large variance)

Task 3.4 Set $\alpha = 1$ and decrease $\beta = 0.1$.

- β = likelihood precision, i.e. how tightly you assume the data pts. should hug the line.
 - smaller $\beta \Rightarrow$ large noise variance $\beta^{-1} \Rightarrow$ each observed y_n is believed to be noisy.
- $p(y_n | w, \beta) = \mathcal{N}(\phi_n^T w, \beta^{-1}) \Rightarrow$ as we decrease β , the bell curve gets wider around every data pt.
- effect on $S_{n \times n}$
 - $S^{-1} = \underbrace{\alpha^{-1}I}_{\text{prior}} + \underbrace{\beta \sum \phi_n \phi_n^T}_{\text{data precision}}$ data contribution to precision is prop. to β , so when we decrease β , we shrink the term $(\beta \sum \phi_n \phi_n^T)$ and makes S looser (posterior weights become uncertain)
 - $m = \beta S^{-1} y$ unless we have a ton of data posterior mean is also pulled towards the prior mean $\Rightarrow$
- overall, the posterior listens more to the prior.

task 3.5 the variance of the predictive distribution $p(y_* | y, x_*)$ contains two components. Describe and relate them to epistemic and aleatoric uncertainty.

- when we ask the model, what y^* might I observe at a new input x^* , it is a normal distribution.

$$p(y_* | y, x_*) = \mathcal{N}(\mu_{y_*}, \sigma_{y_*}^2), \quad \sigma_{y_*}^2 = \underbrace{\phi_*^T S \phi_*}_{\text{model parameter term}} + \underbrace{\beta^{-1}}_{\text{noise term}}$$

- $\phi_*^T S \phi_*$ - the spread is caused by not knowing the exact weights

- we can only get a finite dataset, so the posterior over w has covariance S . Plugging the spread through the basis function vector ϕ_* gives the spread in latent function value $f_* = \phi_*^T w$.

- it shrinks as we get more data ($S \rightarrow 0$ roughly like $\frac{1}{N}$) and if $x^* \in$ well supported region of the design matrix

- epistemic (knowledge driven & reducible)

- β^{-1} - scatter of the observations around the latent function

- even if we know the true weights, each observation is corrupted by $\epsilon \sim \mathcal{N}(0, \beta^{-1})$

- constant $\forall x^*$ as long as β is fixed.

- no additional data can reduce it

aleatoric (inherent, irreducible)

- green band shows epistemic part $\rightarrow$ the uncertainty about the function f^* . Its half width is

$$\pm 1.96 \sqrt{\phi_*^T S \phi_*},$$

and collapses as a posterior over w weights

- yellow band overlays the green one and adds the aleatoric piece. Its half-width is,

$$\pm 1.96 \sqrt{\phi_*^T S \phi_* + \beta^{-1}},$$

even after the green band becomes a hair-line, the yellow band bottoms out at the fixed noise level β^{-1} .

task 3.6 What happens as we get more data, the variance of the posterior distribution $p(y|w)$ of w goes to 0

What happens to variance of

- $P(y_* | y, x_*)$
- $P(y_* | x^*, w)$
- $P(f_* | y, x)$

as N increases? Consider $x_* = 1$. fix $\beta = 1$

- assume, we have calculated
 - the posterior mean of the weights m
 - the posterior covariance of the weights

in code we use function:

Compute_posterior

$$S^{-1} = \alpha I + \beta \mathbb{I} \mathbb{I}^T$$

$$m = \beta S \mathbb{I}^T y$$

Let $\phi_* = \phi(x^*)$ (a column vector of len. D)

- $P(y_* | y, x_*)$ posterior over the observation y^* expected to record at the test point x^* (latent value + noise)

- mean: same as $P(f_* | y, x)$, $\mu_{y_*} = \mu_{f_*}$

- variance: $\sigma_{y_*}^2 = \sigma_{f_*}^2 + \beta^{-1}$

- in code $\text{Var-}y = \text{var-}f + 1/\text{self.beta}$

◦ $p(f_* | y, x)$ posterior over the latent function value f_* at the test point (no noise added!)

◦ mean $\mu_{f_*} = \phi_{f_*}^T m$

◦ variance $\sigma_{f_*}^2 = \phi_{f_*}^T S \phi_{f_*}$

here we assume we know the prior, thus it is deterministic.

◦ $p(y^* | x^*, w)$ the distribution of an observation (if we already know the exact weights. (noise only))

- if handed true w , the only randomness is Gauss. noise ϵ

$$y^* = \phi_{f_*}^T w + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \beta^{-1})$$

$$\text{So, } p(y^* | x^*, w) = \mathcal{N}(\phi_{f_*}^T w, \beta^{-1}) \Rightarrow \mathbb{V}[y^* | w] = \beta^{-1}$$

Task 3.7 Compute prior mean and variance of $f(x^*) = \phi(x^*)^T w = a + bx^*$. Explain why, when $N=0$, the variance

of $f(x^*)$ and of $y(x^*)$ is smallest at $x^*=0$ and increases w/ $|x^*|$.

◦ Compute $\mathbb{E}[f(x^*)] \wedge \mathbb{V}[f(x^*)]$ under the prior $w \sim \mathcal{N}(0, \alpha^{-1} I)$

□ What do we know:

Δ Basis vector: $\phi(x) = [x]$

Δ Prior on w : $w \sim \mathcal{N}(0, \alpha^{-1} I) \Rightarrow \mathbb{E}[w] = 0, \mathbb{V}[w] = \alpha^{-1} I$

Δ Function value: $f(x^*) = \phi(x^*)^T w = a + bx^*$

□ Needed rules

Δ $\mathbb{E}[c^T w] = c^T \mathbb{E}[w]$

Δ $\mathbb{V}[c^T w] = c^T \mathbb{V}[w] c$

◦ Prior mean of $f(x^*)$ ^{constant}

$$\mathbb{E}[f(x^*)] = \mathbb{E}[\phi(x^*)^T w] = \phi(x^*)^T \underbrace{\mathbb{E}[w]}_{\text{Random var.}} = \phi(x^*)^T \cdot 0 = 0$$

◦ Variance of $f(x^*)$

$$\begin{aligned} \mathbb{V}[f(x^*)] &= \mathbb{V}[\phi(x^*)^T w] = \phi(x^*)^T \mathbb{V}[w] \phi(x^*) = \phi(x^*)^T \alpha^{-1} I \phi(x^*) \\ &= \alpha^{-1} \begin{bmatrix} 1 & x^* \end{bmatrix} \begin{bmatrix} 1 \\ x^* \end{bmatrix} \\ &= \alpha^{-1} (1 + (x^*)^2) \end{aligned}$$

- we observe that the variance has two parts:

◦ α^{-1} from the intercept

◦ $\alpha^{-1} (x^*)^2$ from the slope b and increases as $x^* > 0$.

◦ prior predictive variance of $y(x^*)$

→ we add obs. noise variance β^{-1}

$$\mathbb{V}[y(x^*)] = \mathbb{V}[f(x^*)] + \beta^{-1} = \alpha^{-1} (1 + (x^*)^2) + \beta^{-1}$$

◦ $w/N=0$ the posterior = prior

Task 4.1 run the code w/ $\alpha=10$, $\beta=1$. Comment on the quality of fit. Note the values of NLL on training and test set.

- o effect on posterior precision

- $S^{-1} = \alpha I + \beta \tilde{X} \tilde{X}^T$ becomes larger $\Rightarrow S$ gets smaller.

Stronger shrinkage of intercept and linear term.

- o w/ only two basis functions, $[1, x]$, the model can capture trend but **not** the seasonal effect. A very broad prior lets the weights chase the global slope (but gives too extra flexibility to capture month-to-month deviations, which remains in the residuals.